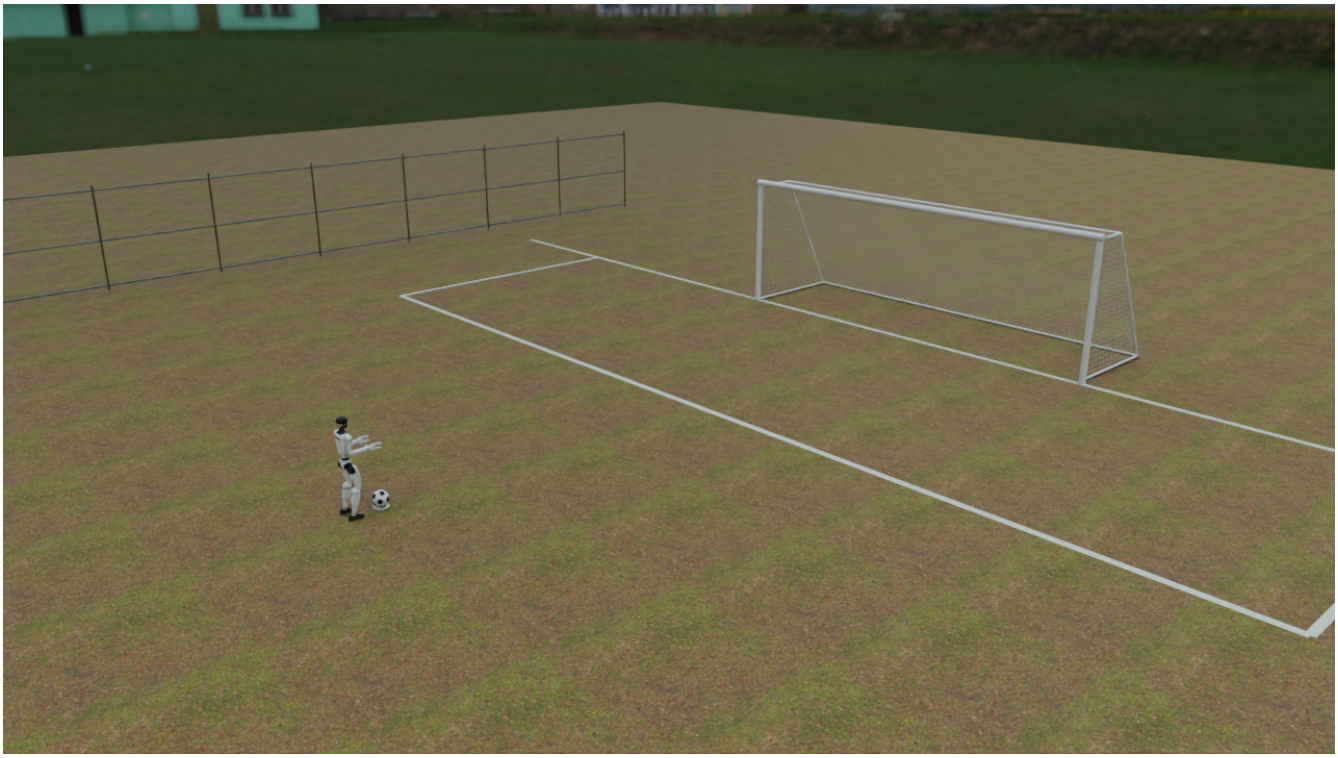
Humanoid football on Newton 1.5

Research, physical implementation and release evidence

Gym Anything for Robotics · Digital environment creation



Newton training-ground overview

This package runs the football environment on Newton 1.5 with researched constitutive models, physical side and roof ball/net interactions and the shared authenticated robot API. Its purpose is to provide a credible digital setting in which another model learns to operate a standard humanoid. Creating an autonomous football solver is not a delivery gate.

The final assisted kick records **5000 uninterrupted physics ticks** and 201 spectator frames at 20 fps. It also renders the actual mounted robot camera at that same rate, retaining selected mounted-camera PNGs, and records a privileged every-tick journal. Its measured wall/simulation ratio is **60.86**, below the limit of 100. The trace reports **true completeness** and **0 evicted samples**. These are measured digital results, not physical specimen measurements.

1. Scope and evidence discipline

This report covers the Newton 1.5 version of the environment.

Three kinds of claims are kept separate: published dimensions/material scales; source-code deductions about force and state ownership; and measured behavior from the recorded Newton runs. Estimated coefficients remain explicit. Physical-world calibration is unavailable and is not used as an excuse for unfinished components. The engineering question is what this digital

model actually does within the task scope, not whether it reproduces an unavailable specimen to laboratory precision.

The public environment contains no external support or outgoing-ball controller. The assisted-kick evidence uses a labelled external pelvis wrench to isolate the coupled leg/ball/pitch behavior. The separate net experiment prescribes an initial ball velocity. Both are valid component fixtures, but neither is presented as an autonomous humanoid goal.

2. Scene, dimensions and physical ownership

Quantity	Implemented value	Evidence/basis
Ball mass	0.430 kg	Mid-range of IFAB 0.410–0.450 kg
Ball circumference	0.690 m	Mid-range of IFAB 0.680–0.700 m
Ball radius	0.1098169 m	Circumference divided by 2 pi
Ball inertia	(2/3) m r² on each axis	Thin-shell reduced model
Goal clear opening	7.32 m × 2.44 m	IFAB Law 1
Main posts/crossbar	0.12 m diameter	IFAB upper allowed width/depth
Penalty mark	11 m before goal plane	IFAB Law 1
Goal depth	2.1 m at ground; 0.82 m roof	Retained plausible training-goal geometry
Net mesh	0.10 m spacing; 4.5 mm cord	Documented 100 mm football mesh; retained cord prior
Gravity	9.80665 m/s²	Standard gravity
Physics sector	x=-34..3 m, y=-12..12 m	Bounded training area, not a whole match stadium

Official laws specify scale, not all material coefficients. See IFAB ball law and IFAB field law. Harrod documents a 100 mm mesh, 4 mm net on its 4G Portagoal. That supports the mesh scale; our 4.5 mm cord, stiffness, mass and tension are retained engineering priors, **not** claimed measurements of that product.

The goal is a **fixed training goal**. Posts and rear/roof/ground support rails have native collision geometry. Cable boundary nodes are fixed ties to the supporting frame; they are not a claim to simulate overturning of an unanchored portable goal. The distant fence, access path, benches and panoramic landscape provide visual context outside the active training sector. They are not additional manipulation tasks or fully collision-modelled scenery.

3. Standard robot and bounded actuation

The G1 is imported from the pinned Unitree URDF, including its inertial definitions, floating base and Dex3 hands. There are 43 commanded revolute joints. The URDF hash is checked before construction; the existing asset registry also identifies the mesh and policy revisions. See Unitree G1 for public specifications.

Joint drives use explicit proportional and derivative gains. The legs retain the original G1 gains; most upper-body joints use 400 N m/rad and 40 N m s/rad, with 20/1 for hand joints. Effort is bounded by the smaller of the URDF effort limit and 120 N m. The 120 N m figure is a cap, not an assertion that every G1 motor can produce that torque. Joint-specific bounds remain available through capabilities metadata.

The control target array uses Newton's coordinate-layout convention explicitly. Unknown joints, non-finite targets and out-of-range target angles are rejected. The API does not silently repair bad commands, teleport feet or stabilize the pelvis. An initial position-hold command can therefore fail to balance a floating humanoid.

4. Newton ownership and multi-rate force coupling

The qualified rigid-body path is Newton **1.5.0**, native CPU MuJoCo **3.11.0**, with full implicit integration, Newton constraint solving, elliptic friction, 64 iterations and a 0.002 s timestep. The package also pins Warp 1.16.0 and MuJoCo-Warp 3.11.0. CUDA dynamics are not advertised as qualified. NVIDIA RTX is used only for rendering. The installed engine's capabilities were inspected before the adapter was built; see Newton 1.5 coupling documentation.

One public tick performs the following sequence:

1. Apply caller joint targets and clear accumulated custom wrenches.

2. Evaluate the pressure-shell shoe contact and aerodynamic wrench from the current state.

3. Subcycle the active cable panels and evaluate pitch contact with the other current ball forces.

4. Sum world-frame force and COM torque; add a labelled fixture wrench only in an experiment.

5. Take exactly one native Newton rigid step; export the resulting state and check finiteness/warnings.

6. Evaluate goal crossing, record the tick and render only if scheduled.

Newton force and velocity arrays are linear/force first, then angular/torque. Public geometry uses XYZ and XYZW quaternions. The retained constitutive view adapter converts to WXYZ as needed. Engine gravity is applied once; the aerodynamic callback excludes its separately reported gravitational term from the applied wrench.

Custom subcycles resolve constitutive contact response; the outer native solver owns the actual rigid state. This is a partitioned coupling, not an all-at-once finite element solve. No contact module assigns the ball's outgoing pose, velocity or spin.

The Newton 1.5 native adapter can fall back from exact shape masks to a coloring approximation on the full G1 collision graph. That fallback admits ball/pitch and ball/right-shoe contacts that should be custom-only. In the full-scene run `assisted-video-03`, with the fallback,

the last two seconds contain 98 overlapping native/custom pitch contact steps and 27.63 mm vertical oscillation. A smaller ball-only world uses exact masks and settles, so ball-only tests do not expose the problem.

The repository adapter therefore reserves one unused MuJoCo collision type bit for exact ball partners, also updating body-level broad-phase masks. It checks every non-ball mask pair remains unchanged and fails construction if exact ownership cannot be established. Posts, other robot links and the custom pitch/shoe laws remain enabled under their intended owners. No material parameter is changed to conceal the bounce. See MuJoCo collision filtering and `collision_ownership.py` for the implementation.

5. Pressure-shell shoe impact

The right shoe is a deterministic convex hull of the pinned visible G1 shoe mesh: 1,248 proxy vertices and 2,492 triangles. The closest surface point and normal define the contact patch. It is not an arbitrary box; the convex hull does fill concavities. The original URDF's native collision geometry remains responsible for other links.

For indentation d and ball radius r, the retained reduced law uses:

```
Vcap(d) = pi d² (r - d/3)
V(d) = V0 - a Vcap(d),  a = 0.22
pabs(d) = pabs,0 (V0 / V(d))^1.40
Fpressure = (pabs - patm) a pi d (2r - d)
Felastic = Fpressure + Kshell d^2.30
Fn = max(0, Felastic + c(d, meffective) d_dot)
```

Initial gauge pressure is 80 kPa. The shell coefficient is derived from the prior reference point 2,926 N at 62 mm compression, after subtracting pressure force. This is a retained impact-scale prior informed by the literature review, not a new fit to hardware. The damping ratio is 0.20. Tangential displacement is a state variable with 18,000 N/m stiffness, 7 N s/m damping and static/dynamic friction 0.62/0.48. Slip updates remove stored tangential spring energy rather than retaining an inconsistent force-clamped spring.

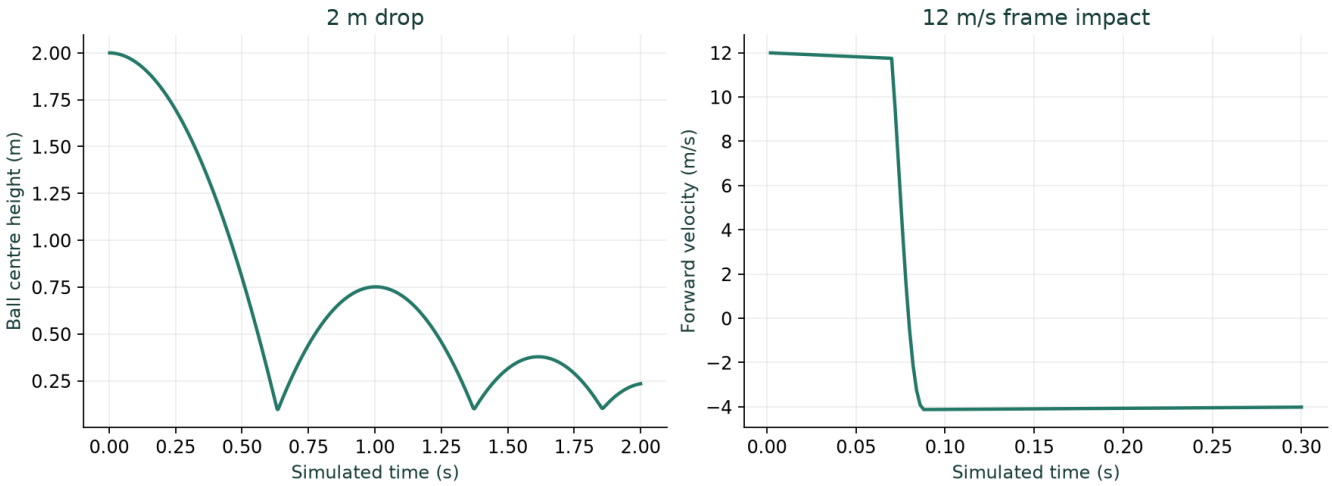
The response tensor uses the live articulated mass matrix and contact-point Jacobian: W = J M^-1 J^T. A contact-conditioned planted-support projection is used only when the other foot has a nearby native ground contact; this is an ideal-support approximation, explicitly reported beside the free tensor. See Khatib's operational-space formulation. Football impact time/deformation scales are supported by Shinkai et al..

The coupling reports equal/opposite force and impulse closure, contact work, constitutive dissipation and effective mass. These numerical ledgers establish accounting consistency; the actual kick video and force trace establish that the implemented robot/ball interaction occurs. Neither is mislabelled laboratory validation.

6. Pitch, rolling and frame contact

The ball/pitch pair uses the retained compliant law, not the rigid plane restitution: Fn = max(0, 250000 d^1.5 + c(d) d_dot). The normal damping ratio is 0.26; static/dynamic friction is 0.58/0.44; tangential stiffness/damping is 14,000 N/m and 5 N s/m. Rolling-resistance coefficient 0.130 and spin-resistance coefficient 0.010 represent the aggregate grass/root-zone system, not individual grass-blade friction. The pile direction adds an 8% friction anisotropy.

These priors were initialized against public pitch performance scales in the retained research, including FIFA's natural-surface testing programme. The 2 m drop fixture measures a first rebound height of **0.751 m**. The plot shows the entire actual height trace; the pass criterion is a bounded rebound above 0.4 m, not a declaration of FIFA certification.



Drop and native frame impact

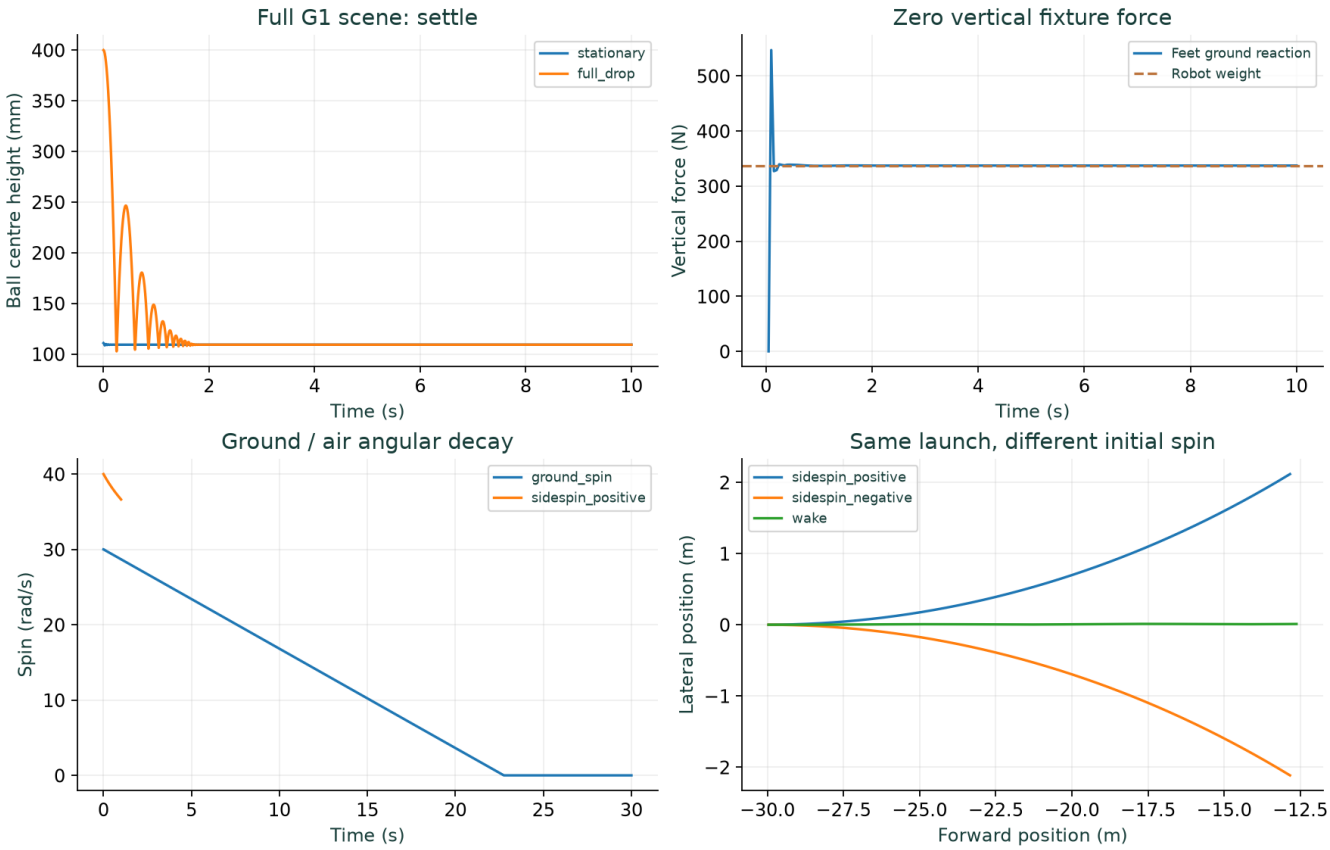
Robot/ground and ball/frame contact remain native MuJoCo constraints. In the 12 m/s frame fixture the minimum forward ball velocity is **-4.123 m/s**. The reversal demonstrates native collision.

The full-G1 stationary and 0.4 m drop fixtures run for 10 seconds each, checking the final two seconds rather than stopping after one rebound. Sliding runs for eight seconds and vertical spin for 30 seconds. The measured settling/support results are:

Actual fixture	Final-two-second height range (mm)	Final speed (m/s)	Final spin (rad/s)
stationary	0.00000	0.000009	0.000513
full_drop	0.00000	0.000005	0.000383
slide	0.00000	0.000002	0.000416
ground_spin	0.00000	0.000010	0.000521

Stationary robot weight: 337.292 N; mean foot reaction: 337.293 N. Both feet remain load-bearing; pelvis height varies 0.571 mm over seconds 8–10. Vertical fixture force: 0.0 N.

No active native contact with a custom-owned ball pair is allowed in these checks. The grounded G1 fixture applies horizontal/orientation assistance but **zero vertical force**: measured foot reactions carry its weight. It is not an autonomous balance claim. The initial roughly 1 cm sole clearance is a short gravity-settling transient, not a permanently floating standing pose.



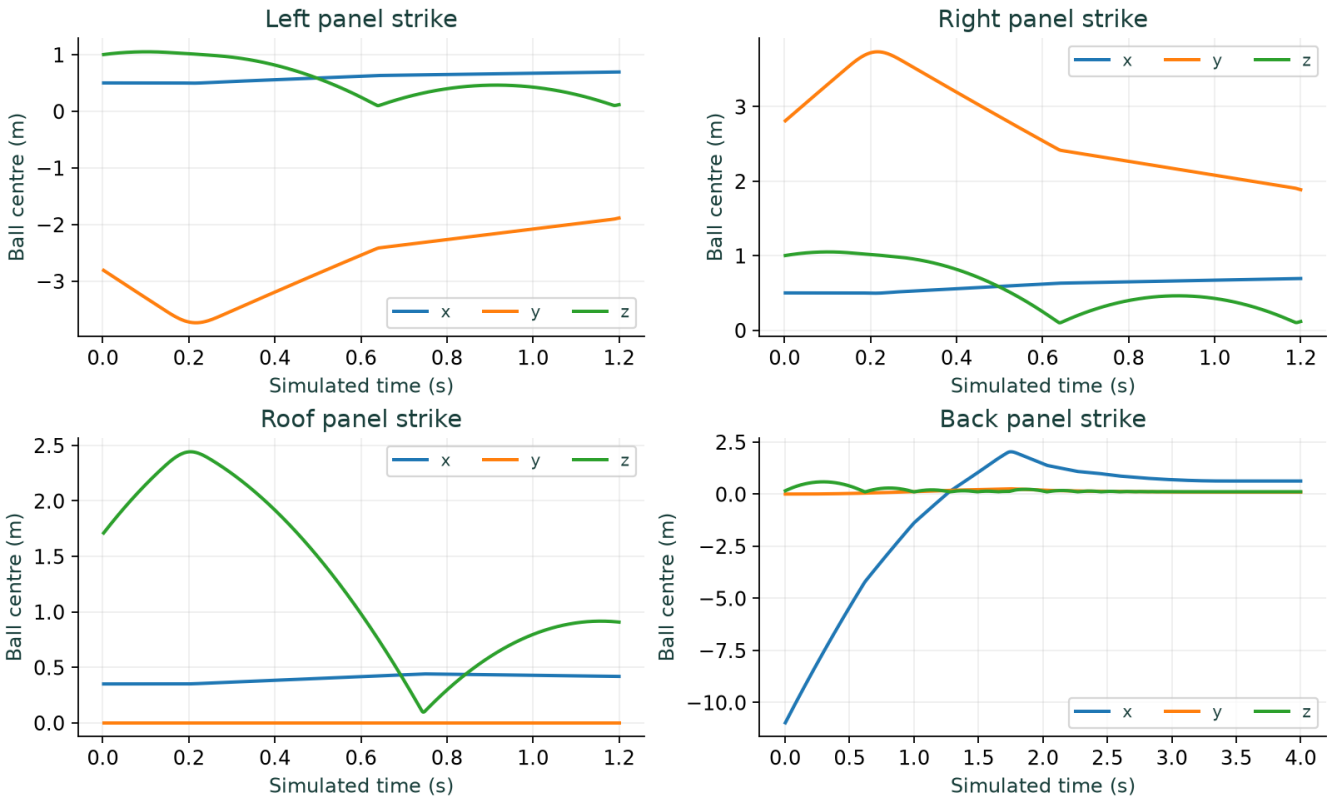
Measured settling, supported weight, spin decay and opposite-spin flight

7. Four-panel deformable ball/net enclosure

Each rendered cord is an edge in a mass-node cable graph. The current enclosure has 2,072 back nodes, 432 on each side and 666 on the roof. A 0.2 s, 24 kHz initialization settles the panels before the episode; its parameter-hashed cache is separate from the episode clock. Mesh spacing is 100 mm, smaller than the ball diameter.

Edge force is tension-only: T = max(0, T0 + k (length-rest_length) + c length_rate). The retained pretension is 6 N and stiffness is 2,500 N/m. The field is named `axial_stiffness_n`, but the executed formula multiplies extension in metres, so its effective unit is N/m. Cord linear density is 0.006 kg/m. Ball/cord contact uses a 1.5 exponent, stiffness 180,000 N/m^1.5, damping ratio 0.22 and dynamic friction 0.24.

The ball drives node motion and receives the reaction wrench. Anchors carry support reaction. Separate actual strikes confirm contact with left, right, roof and back panels, with finite state and no native numerical warnings. The back launch reaches the goal and deforms the net by **0.1119 m**, with a peak reported contact force of **46.15 N**.



Physical net strike trajectories

A panel sleeps after its post-contact integration window and wakes on renewed ball proximity, so the enclosure does not stay inactive after the first strike. Full long-time cloth flutter and robot/rope contact are not simulated. These are ball-catching cable panels, not general-purpose clothing or a net-manipulation task. Numerical tests of ball strikes must not be generalized to robot entanglement.

A targeted repeat-contact rollout launched the ball into the back net, then applied a labelled +8 N forward ball force from 4.0 to 4.5 s. Contact-bearing outer steps increased from 58 to 116 without restoring or assigning ball state. This verifies re-activation on a second contact; it is not an autonomous robot shot.



Separate launched-ball fixture, after back-net contact at 1.95 seconds

8. Flight and aerodynamic priors

Air-relative velocity gives Re = rho |v| (2r) / mu and spin parameter S = r |omega|/|v|. The aerodynamic model retains its Reynolds-dependent drag transition (Cd 0.50 to 0.20 around Re 250,000), bounded Magnus term, seam orientation dependence, spin decay, buoyancy and coherent low-spin wake state. Air is initialized at 293.15 K, 101,325 Pa and 50% relative humidity. Drag acts against relative velocity and spin torque opposes spin.

These are researched reduced models, not a fluid simulation. The existing source catalogue contains the relevant experimental literature, including Carre et al. on seams, Hong et al. on spinning balls and Hong and Asai on knuckle-ball forces. Those equations are not replaced to improve a score. One-second airborne fixtures compare +40/-40 rad/s sidespin and topspin/backspin from the same position (-30,0,3) m and velocity (20,0,7) m/s. Sidespin ends at lateral positions **2.115, -2.115 m**; topspin/backspin end at heights **2.531, 6.779 m**. Each retains spin decay, and successive physical quaternion changes agree with angular velocity within the recorded tolerance. A zero-spin run exercises the time-varying wake. These are integrated trajectories, not just a force-function assertion or a 0.2-second finiteness test. Panel meshes are attached to the body's same XYZW transform, so visual rotation follows physical orientation.

The adapter initializes both `body_qd` and the free-joint velocity before `eval_fk`, so forward kinematics cannot overwrite the initial body velocity with an uninitialized value. The launch evidence records the requested initial velocity (12,0,3) m/s and spin (0,0,10) rad/s.

9. Whole-ball scoring and compliant ground

The plane test in `football_task.py` checks the **trailing ball surface**, not just its centre. It linearly interpolates the crossing between consecutive physics ticks. Side clearance uses half goal width minus |y| minus ball radius; upper clearance uses goal height minus z minus radius. No score is awarded for merely touching the net.

Pitch compression matters at the line. In a ground launch, the ball's nominal lower surface at crossing is 3.902 mm below z=0 because the pitch is compressed, more than a 2 mm geometric margin allows. The verifier therefore admits current simulated pitch compression **only while the pitch supplies positive normal force**, in addition to the 2 mm geometric margin. It does not enlarge the side or upper opening.

Targeted checks accept a compliant supported ground crossing and an ordinary inside crossing, and reject a below-ground unsupported ball, a deeper below-floor trajectory, an outside-post crossing and an above-bar crossing. The actual launched-ball fixture scores under this plane test: **true**. The restricted policy does not receive this privileged ground-truth evaluator through its observation channel.

Task verifier and reward

The football/goal task scores with `task.py`. Its rules and thresholds are documented in the module docstring. The plane test above remains in `football_task.py`.

- **Opening at the frame.** In or out is decided from the ball centre where it crosses the plane of the post and crossbar axes (x = 0), against those axes (|y| < 3.72 m, z < 2.50 m). The frame is solid there, so the verdict cannot depend on how far the side or roof net behind the line has yielded. A trailing-surface test at x = 0 would put the boundary on the net planes: robot-free launches 7-21 mm inside a post or 10 mm under the bar, touching neither, would be latched "outside" permanently.

- **Back edge of the line.** The whole ball is over the goal line when its trailing surface passes x = +0.06 m, the back edge of the 0.12 m drawn line and of the posts.

- **Kick, not handball.** In the robot world, any arm or hand touch on the ball ends the attempt; a goal counts only if the robot's last touch before the line was a leg touch (the hip links, i.e. hip-joint housing and thigh, the knee/shin, ankle or foot). The G1's shoulder-joint housings (`shoulder_pitch_link`, `shoulder_roll_link`) sit above the upper arm, i.e. above IFAB Law 12's armpit boundary, and count as trunk: a touch there caps progress like a chest touch instead of ending the attempt. The upper arm from `shoulder_yaw_link` down, the elbow, wrist and hand remain arm. Robot-free fixtures have no kick rule.

- **Continuous score.** 1 on a goal, else 0.9 x progress, progress being how much nearer the goal window the ball is than at the episode start, frozen when the attempt ends.

The compliant-ground floor allowance above also applies here. The recorded crossing margins (`goal_crossing.lateral/upper/lower_clearance_m`) are measured to the decision boundaries at the frame plane (3.72 - |y|, 2.50 - z, z - r + floor allowance), so they are positive exactly when the ball passed inside. Failures are credited by where the ball is when the attempt ends or the score is read, so a post rebound is credited where it lies (`closest_approach_m` records the near thing). The owner's per-action reward is the current score level, not an increment. An incidental

arm or hand touch earlier in the play still ends the attempt. This is an accepted, documented limitation.

The leg rule reads "kick" as play with any leg link, deliberately wider than the IFAB glossary's foot-and/or-ankle definition: on the G1 the shin meets the ball with the shoe in an instep strike, and a shot in off the shin or thigh is still the kick asked for. Accepted consequences: a ball carried between the knees or pushed along by the shins into the goal scores, and because the hip-joint housing is a leg link a bump off the side of the hip counts while the same bump on the pelvis does not.

10. Camera and scene rendering

The scene uses the same body poses and net-node positions as physics. The render worker receives a sequence number and tick, updates USD transforms, and returns the exact tick with `physics_steps=0`. It contains no rigid-body physics scene and never advances a PhysX timeline. A stale sequence or render-induced tick is rejected.

The mounted camera uses the pinned URDF D435 location/orientation, a 69° horizontal FOV and a virtual filmback. It is an ideal RGB observation, not a calibrated sensor noise/depth emulator. Teacher viewpoints include overview, sideline, contact and goal. Restricted clients cannot use those free spectator views.



Right shoe and ball at actual scale

The visual model includes smooth standard robot meshes, spherical panel geometry, contact-owned reduced shell deformation, physical net cords and rails, field markings, an outdoor HDRI and a PBR ground material with correctly scaled UVs. Render-only grass blades add subgrid relief; the pitch law owns homogenized compliance. The surface is an autumn community training ground, not a pristine stadium or a photorealistic turf claim.

Appearance assets are CC0: Suburban Football Field by Grzegorz Wronkowski and Leafy Grass by Charlotte Baglioni. Downloads are content checked. The HDRI's distant structures are background imagery, not objects the robot can physically interact with.

With texture streaming enabled, the first observation shows a coarse turf mip even after shader warmup. The render worker therefore disables texture streaming **before** stage loading and requests synchronous asset loads. This follows the documented RTX streaming setting. Two cold-start contact-camera captures at the same tick have identical static turf pixels; the first image was visually inspected for full texture detail. The videos are rendered with this setting, not patched with frames from another simulation rollout.

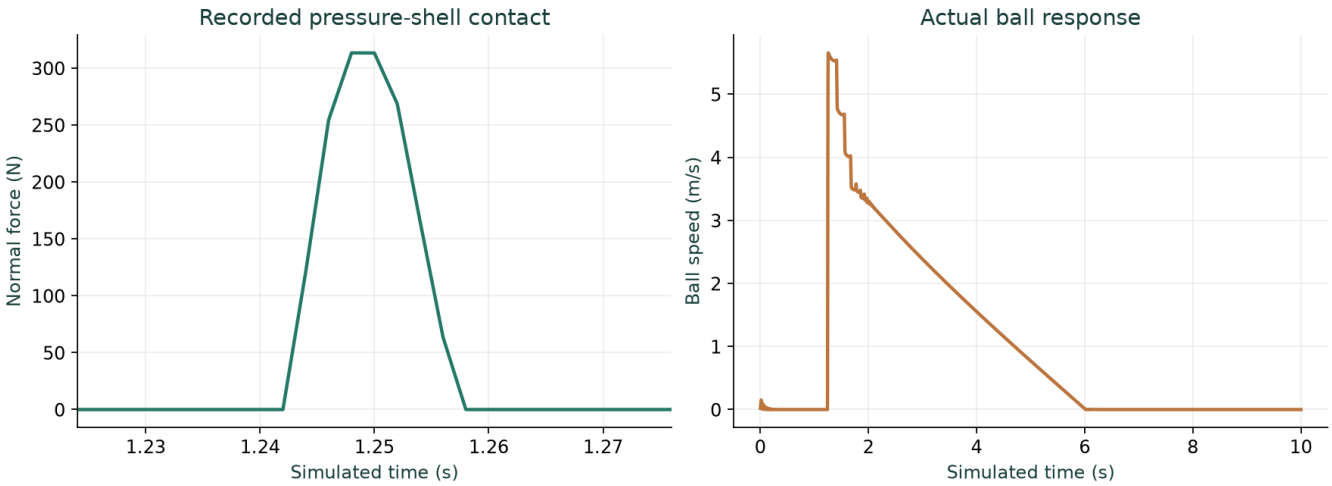
11. Assisted evidence and temporal continuity



Close-up assisted motor-driven kick at 1.25 seconds; independent uninterrupted fixture

The support fixture applies a recorded pelvis wrench, not a weld or state assignment: F = 1200 (p_target-p) - 100 v + m g e_z; torque = 200 rotation_error - 20 omega. This is explicitly a **suspension rig** carrying the robot's weight. Its approximately centimetre-scale sole clearance must not be presented as grounded standing. The separate stationary recording uses Fz=0 and verifies load-bearing foot contact. Removing suspension while replaying the same kick targets missed the ball; this is recorded as an unsuitable fixture motion, not a physics failure or a solved policy. The leg receives bounded quintic joint targets: windup, a 180 ms stroke and recovery. The pressure-shell shoe model generates ball motion. The run's first contact is at **1.242466 s**, peak foot force **313.33 N**, and impulse on the ball **(2.4087, 0.1520, 0.3692) N s**. Peak external support force/torque are reported in its manifest.

No reset, checkpoint restore or outgoing-ball assignment occurs after tick zero. Each MP4 has its own run ID, frame/tick list and SHA-256 checksum. The wide kick and the launched-ball goal are separate, clearly labelled experiments; they are never spliced into a supposed autonomous solve.



Measured contact force and ball response, sampled every outer physics tick

The video exporter reads and encodes actual rendered frames. It records every 25th tick, including both endpoints. At 20 fps a 10.0 s rollout therefore produces a 10.05 s file; this is the endpoint's final 50 ms display time, not extra simulation. Playback and selected frames must be inspected, not inferred from a zero exit code.

12. Privileged history, checkpointing and recovery

The shared HTTP gateway writes before/after action records with request IDs, action IDs and UTC timestamps. Its teacher-only trace contains every 500 Hz outer physics state, joint targets, all native contact witnesses, constitutive summaries and every net node's position and velocity. Native contacts are stored column-wise, with lossless Zstandard compression, preserving exact numeric bytes.

MuJoCo includes inactive proximity witnesses within its margin. The journal keeps them, but contact begin/end events use active constraint addresses rather than mistaking every nearby pair for a physical impact. Geometry ordering is not sorted away: the recorded world force acts on geom2 as labelled.

This is **not a 24 kHz microstep archive**. Constitutive microsteps are represented by impulse, peak force and energy summaries. Native force witnesses belong to the start of the native step; poses belong to its end. The teacher can locate events at 2 ms resolution without being promised unavailable sub-microstep ordering.

Storage is bounded, with explicit missing ranges, evictions and recording errors. A diagnostic video with a 512 MiB budget captured 5,000 samples but evicted 2,695. That trace is not called complete. The kick video uses a 2 GiB budget and the lossless columnar layout. Its actual completeness appears on this report's cover. Disk reserve is checked before actions and during recording. Recording failure is not silently ignored: it prevents or stops physics and requires owner recovery.

Checkpoints contain both Newton buffers, native integration/warmstart state, targets, ball contact histories, wake state, cable positions/velocities and goal state. Continuation after a checkpoint saved during shoe contact reproduced the next 100 ticks with maximum pose, velocity and net-position error **0, 0, 0**. Restoring still starts a new continuity generation. Byte-exact continuation is not permission to splice restored episodes into a continuous-video claim.

13. API integration and the robot/teacher boundary

`gar.create("football", robot="g1")` creates one persistent instance. It supports observations, named-joint actions, state, reset, render, geometry and checkpoints. Idle time does not advance physics. The caller may command an empty target dictionary to advance under the current drive targets. There is no direct-torque or phantom-force policy action. The fixture API is local experiment code, not a remote policy privilege.

The two-key HTTP contract is documented by `GET /v1/help`. Restricted clients obtain mounted RGB, joint encoders, robot limits and named joint commands. They do not obtain object poses, full meshes, spectator views, history, checkpoints, resets or the hidden goal evaluator. Teacher credentials can retrieve those diagnostics. Tokens are not committed with evidence. Use a private network/TLS proxy for external access.

Actual local HTTP tests exercised help, capabilities, observation, action, state, geometry, per-action traces and forbidden routes. A labelled fault injection verified `ready=false` and owner reset recovery. Unknown-joint rejection advanced zero ticks. A separate 30-tick checkpoint continuation also had zero pose and velocity error. The camera-enabled HTTP run passes: **true**.

14. Measured performance and qualification table

Workload	Simulated seconds	Wall/sim ratio	Result
drop	2.00	1.09	pass
frame	0.30	0.98	pass
free_flight	0.20	0.96	pass
net_left	1.20	9.01	pass
net_right	1.20	8.28	pass
net_roof	1.20	9.22	pass
net_back	4.00	7.60	pass
Full G1, two RGB views, every-tick history	10.00	60.86	pass
Goal/net video, full G1 and history	4.00	70.38	pass
Shoe-contact close-up and history	3.00	64.26	pass
Grounded stationary G1 and history	10.00	62.22	pass
Ground spin and history	5.00	61.65	pass

The final camera workload is batch **one**, full G1 and enclosure, 500 Hz physics, 20/24 kHz constitutive subcycles, every-tick history and two 1280×720 RGB views at 20 fps with 16 path-tracing samples. Reported ratios are wall seconds divided by simulated seconds.

Renderer/world startup is excluded from the timed rollout and must not be mistaken for free startup. No batch sharding or aggregate-throughput claim is used. Other bounded qualification jobs briefly shared the machine.

The table uses runs with exact collision ownership, including their history/render costs.

The renderer is not fast enough to claim real-time gameplay. The acceptance threshold here is below 100, not 1. No timestep, resolution or sample-rate reduction was used to pass the final workload. The principal practical costs are image rendering/readback, per-tick history and constitutive contact subcycling, rather than an unbounded video cache.

15. Warnings, failures and model limits

An initialization warning says MuJoCo-Warp MULTICCD has at most one contact for certain cylinder/box/mesh pairs. It originates while Newton prepares the optional Warp model; the qualified execution path calls native CPU `mj_step`. It was investigated, not suppressed. Native numerical-warning counters were zero in the physical qualification runs. The warning does not prove every contact manifold is exact; the tested frame impact and falling/assisted robot behavior are the actual evidence.

Renderer warnings were reviewed separately: installed-extension deprecations, previous crash-dump notices, PCIe/readback performance warnings, and regular-texture fallback after deliberately disabling streaming remain visible. No renderer `[Error]` entry was found in the final run logs. Actual captures and full-workload timings, rather than suppression of vendor messages, qualify this path.

Known limits are concrete: CPU-only qualified dynamics; ideal RGB; approximate thin-shell ball and pressure-contact mode rather than volumetric FEM; a convex right shoe proxy; native rather than pressure-shell contact for other robot links; partitioned coupling; ball-only cable contact, fixed anchors and post-contact net sleeping; bounded training sector and visual-only distant context. Friction and material numbers are research-backed priors, not named-product measurements. There is no new variation distribution and no claimed autonomous Newton solver.

The curl controller of the earlier Isaac version, transferred directly to Newton, does **not** reproduce its goal. It remains finite but the robot falls, reaching a minimum upright cosine of -0.610; the ball's maximum speed is 2.309 m/s. No physics coefficient is tuned to force the old trace. A finite run is evidence for its tested actions and duration, not a universal guarantee that any adversarial joint sequence can never destabilize a numerical simulator.

16. Reproduction

Install the release and start the penalty task with `gar run football --task goal --port 8080`. The required runtime versions are pinned by the release requirements. No assets are fetched during a physics tick.

Each recorded run keeps its scene images, continuous MP4 and manifest, compact physical traces and API/coupling results. The PDF is generated from the recorded JSON/NPZ evidence. Tests, plots and videos support different claims; none is substituted for the others.

17. Source and asset provenance

This report relies on the official IFAB/Unitree/Newton sources, the Harrod mesh specification, the retained peer-reviewed impact/aerodynamic studies and the asset sources linked above. Publication-based priors are distinguished from manufacturer dimensions throughout this report.

Pinned G1 URDF SHA-256: `555edb931dbe8780a82c778f7c7f53fe7f4f6d73c7e791845631cda9e8042302`. Pinned right-shoe convex proxy SHA-256: `232a588bef5c4364daa573c77f6f78874f17438f7a6f2cbf2efbd6d1b46e80fd`. Exact source and downloaded appearance checksums are recorded by the package. Poly Haven assets use CC0; Unitree source assets retain their repository license notices. The robot is not a custom easier surrogate.

The conclusion is bounded but useful: this is a working, inspectable Newton football environment with physically driven robot/ball behavior, a ball-catching cable enclosure, controlled information access and reproducible evidence. It is not a renamed video, an autonomous-policy success, or a requirement to perform unavailable real-world calibration.